

ZHAOLI CAO

www.erictsao.cn | tsaochaoli@gmail.com | +852 54282733

EDUCATION

The University of Hong Kong

Master of Finance in Financial Technology | GPA: 3.99/4.0

Hong Kong, China

2025.09 – 2026.11

- **Honors:** BOC Talent Cultivation Scholarship HK\$100,000, Fintech Scholarship, Vice President of the HKU Elite Club

Fudan University

Bachelor of Science in Mathematics (Minor: Economics & Finance) | GPA: 3.76/4.0

Shanghai, China

2020.09 – 2025.06

- **Courses:** Stochastic Calculus, Measure Theory, Real Analysis, Mathematical Statistics, Numerical Optimization, Time Series

- **Honors:** WorldQuant Challenge Gold Medal (2025), First-Class Scholarship of Fudan University(2024), IMC Prosperity3 Global Top 2% (2025), Kaggle DRW Crypto Market Prediction Top 10% (2025), MCM Meritorious Winner (2023)

PROFESSIONAL EXPERIENCE

NonConvex Technology, Quantitative Researcher(HFT: 35B average daily volume) Shanghai, Mar 2026 – Present

Conducted proprietary HFT research on equity index futures, focusing on microstructure alpha and execution optimization

- **RL Execution:** Built D3QN agent on L10 LOB + α -signal states for IF futures, reducing shortfall by 0.47 bps vs VWAP. ($t = 3.1$, evaluated over 60+ trading days).
- **Rule-Based Strategy Optimization:** Identified forecast-execution latency mismatch in multi-horizon signals; designed signal-strength gating, adaptive exit logic, and spread-conditioned entry thresholds. OOS Sharpe 2.12→5.35
- **Hierarchical RL:** Two-tier D3QN — minute-level router over K=3 regime sub-policies (K-means on volatility/OFI/spread) with switching-cost regularization, lifting net intraday alpha 1.12 bps across IF/IH/IM.

Matrixport, Quantitative Trader (AUM: \$10B+ in digital assets)

Hong Kong, Sep 2025 – Dec 2025

Conducted crypto microstructure research, focusing on L2 signal extraction and multi-frequency factor modeling.

- **Market Making Strategy Research (Crypto):** Studied inventory-aware optimal quoting in a limit order book, adjusting bid-ask spreads and order skew using L2 order book dynamics and funding-rate signals to improve execution quality
- **Microstructure & Regime Detection:** Based on L2 order book and funding rate data, extracted market microstructure features and employed a Hidden Markov Model to identify latent market states, enabling dynamic signal adaptation.
- **Signal Integration:** Tested whether NLP sentiment adds value beyond L2 microstructure; marginal Sharpe uplift ~0.3, limited in high-frequency regimes.

Geek Union Fund, Quantitative Research

Shanghai, Jun 2025 – Aug 2025

Responsible for factor mining on daily and hourly CTA data, as well as multi-window factor training and portfolio construction.

- **Factor Mining:** Conducted daily/hourly CTA factor research via multi-window training, dynamic smoothing, and IC-return filtering to build a library of 80+ signals. Applied regime-sensitive sampling to improve robustness across market states
- **Operator Development:** Refactored core signal modules, achieving ~77% runtime improvement via Numba JIT and pybind11-based C++ extensions with SIMD (AVX2). Integrated multi-horizon operators, cross-sectional allocation, and risk-budget weighting to speed up tuning and portfolio construction
- **Loss & Statistical Regularization:** Designed a robust loss (winsorization, Huber, directional penalties) and applied Newey–West adjustments to address heteroskedasticity and autocorrelation in factor evaluation
- **Rule-Based Features & Portfolio:** Extracted rule-based K-line features (IC up to 0.12) and built multi-factor portfolios using SGD-based allocation + KMeans, achieving Sharpe 2.64 / Calmar 4.15 out-of-sample

KR Capital, Quantitative Research

Shanghai, Jan 2025 – May 2025

Responsible for CTA strategy and stock strategy research, backtesting, and deployment in simulation and live trading

- **Cross-Sectional Futures Alpha:** Built minute-level multi-factor long-short strategies using roll yield, basis momentum, and other high-frequency features. Applied XGBoost and Newey–West residual adjustments, achieving Sharpe 1.78, mdd 9.6%
- **High-Frequency Statistical Arbitrage:** Built a latency-aware stat arb strategy with dynamic hedge ratios (rolling GLS), accounting for execution frictions and validated out-of-sample, achieving Sharpe 3.45 after accounting for transaction costs

PROJECT EXPERIENCE

Self-Built End-to-End Quant Framework (A-share / Crypto Capstone Project)

Mar 2026

Built prod-grade system (Rust FE + Polars + Numba JIT), signals cross-validated vs. industry benchmarks (JointQuant)

- PIT (announce_date) + clock-truncated backtest, stateful matcher, Barra CNE6 (WLS + NW + LW shrinkage), 2-layer risk
- Systematic factor generation via 36-operator DSL with multi-stage filtering; eval pipeline shared with GP/RL search.
- Processed 1,000+ broker reports (200K tokens/full text) into 20+ structured features feeding AlphaAgent, zero look-ahead

IMC Trading Prosperity3 High-Frequency Trading Competition (Global Top 2%)

Apr 2025

Served as team leader, overseeing case studies and strategy development; led quantitative strategy design and manual trading

- Strategies included: order book market making (spread & inventory control), ETF premium arbitrage (intra-day dislocations), options mispricing via Black–Scholes/IV bias, cross-asset relative value, and signal-based momentum following.

SKILLS

Programming: Rust(proficient), C++ (advanced: STL, RAIL), Python (advanced: NumPy, Pandas, Pytorch, Polars), Linux, R

Language: Chinese (native), English (fluent, TOEFL: 109 (Reading: 30; Listening: 29; Speaking: 25; Writing: 25))

曹肇立

www.erictsao.cn | tsaochaoli@gmail.com | +86 15301892244

教育背景

香港大学，金融科技 | GPA(3.99/4.0)

香港, 2025.09 - 2026.11

· 荣誉: 中银香港人才培育奖学金 HK\$100,000, 金融科技择优奖学金, 香港大学精英协会金融科技副会长

复旦大学，数学与应用数学（辅修专业：经济学&金融学） | GPA(3.76/4.0)

上海, 2020.09 - 2025.06

· 荣誉: WorldQuant金牌(2025)、Kaggle DRW Crypto Market Prediction 全球前10%(2025)、高频交易比赛IMC Prosperity全球前2%(2025)、美赛M奖(2023)、复旦大学专业奖学金

实习经历

非凸科技，高频量化研究（自营交易，日均成交约350亿元）

上海, 2026.03 - 至今

负责股指期货高频策略研究，涵盖微观结构信号挖掘、执行算法优化及低延迟交易系统开发

- 强化学习执行：基于 L10 盘口 + α 信号训练 D3QN 智能体，用于 IF 执行，相比 VWAP 降低实施落差 0.47 bps
- 规则策略：引入信号强度门控、自适应止出及价差条件化入场机制，样本外 Sharpe 由 2.12 提升至 5.35
- 分层强化学习：两层 D3QN(routerK=3 专家, K-means regime+切换成本)，在 IF/IH/IM 上提升日内净 alpha 1.12 bps

Matrixport，量化交易（数字资产管理规模：100亿美元以上）

香港, 2025.09 - 2025.12

负责Crypto市场的量化研究与交易，高频因子挖掘、验证与优化

- 做市策略建模：围绕限价订单簿下的库存约束做市问题，研究inventory-aware最优报价机制，引入成交强度与库存惩罚项建模，结合 L2 订单簿动态与资金费率信号，对bid-ask spread与挂单倾斜进行自适应调整
- 微观结构与状态识别：基于 L2 订单簿与资金费率数据提取市场微观结构特征，采用隐马尔可夫模型HMM识别潜在市场状态，按状态后验概率对进出场阈值与因子暴露进行加权调整，策略 Sharpe 提升约 23%
- 多模态时序建模：构建端到端策略，采用双塔结构与门控融合机制，结合状态条件化与表示对齐优化，融合 L2 订单簿序列与 NLP 情绪特征，通过 LSTM / Transformer 学习状态条件化信号，测试集 Sharpe 2.77

积优基金，量化研究

上海, 2025.06 - 2025.08

CTA日频及小时频因子挖掘、因子多窗口期训练及组合

- 单因子训练：多窗口训练，结合IC-收益联合筛选80+因子，并以状态分层建模检验异质市场下的稳健性
- 损失函数及统计修正：设计自定义损失函数，结合95分位数winsorize、Huber Loss与方向一致性惩罚，增强异常值鲁棒性与信号稳健性；在因子评估中引入Newey-West回归修正，应对异方差与自相关问题
- 规则类因子挖掘：基于K线模式提取规则类特征，单因子IC值0.12
- 框架优化：重构模块，基于Numba, PyTorch与SIMD(AVX2)实现77%性能提升并引入多周期预测&风险预算

KR Capital，量化研究

上海, 2025.01 - 2025.05

负责期货截面策略和股票量化策略的研究回测及模拟盘实盘部署

- 期货截面Alpha：分钟级多空组合(期限结构/基差)，XGBoost+滚动筛选(NW修正)，Sharpe 1.78 / MDD 9.6%
- 全球宏观轮动策略：利用GMM+EM聚类划分市场状态并动态调整ETF配置，实盘部署至今Sharpe 2.35

项目经历

自建全流程量化交易框架（A股/加密货币，8万行Capstone项目）

上海, 2026.03

基于Rust内核因子引擎+Polars+Numba JIT构建，全链路信号与聚宽/RQAlpha交叉验证对齐

- 结果对齐：PIT 回测（公告日 + 时钟截断）、撮合器（成交/冲击/容量）、Barra CNE6 + 双层风控，严格复现
- LLM 因子挖掘：36 算子 DSL，假说 → 表达式 → 多维打分 → 三阶晋升关卡，与 GP / RL 共享评估管道
- SSRN复现：单篇200k tokens复现1000+ 篇研报及论文，抽取 20+ 维特征送入AlphaAgent, 提取1000+独立因子

IMC Prosperity3高频交易竞赛（全球前2%）

上海, 2025.04

担任队长，带领团队学习往年案例，分析数据，构建并优化策略，并负责开发量化策略及所有手动交易

- 构建基于订单簿深度与价格波动的做市报价模型、利用ETF市场价与组件净值间价差的跨品种套利机制、结合Black-Scholes期权定价与隐含波动率偏差的定价套利模型、基于协整与价差Z-score构造的跨资产价差回归策略

基于机器学习的比特币价格预测与量化交易策略研究（优秀毕业论文）

上海, 2025.02 - 2025.05

构建比特币价格预测模型，并设计优化的量化交易策略

- 构建多模型比特币预测与量化策略，引入特征工程与贝叶斯优化，结合滚动回测与成本建模，Sharpe 2.70

个人技能

编程语言: Rust (熟练), C++ (精通: STL, RAI), Python (精通: NumPy, Pandas, SciPy, Pytorch), Linux, R, Matlab

语言能力: 中文(母语)、英文(流利, TOEFL: 109; 阅读: 30, 听力: 29, 口语: 25, 写作: 25)